

Mini Meeting

31 Aug 2026

champ



2.81 Ltr

upper band=30; lower band 20 cm

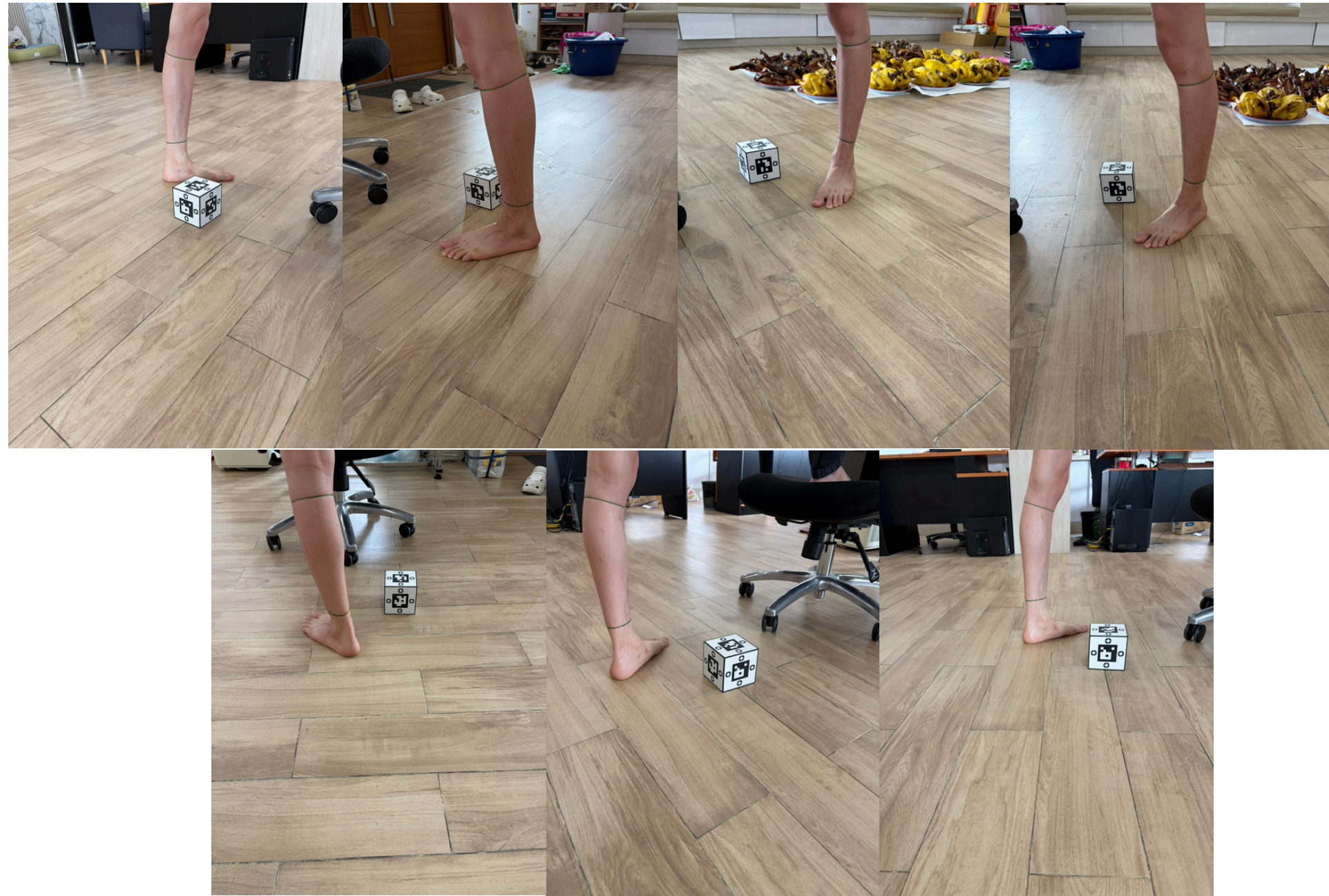
sunshine



3.13 Ltr

upper band=31; lower band 22 cm

keng



2.21 Ltr

upper band=29.5; lower band 19.5 cm

orange shirt

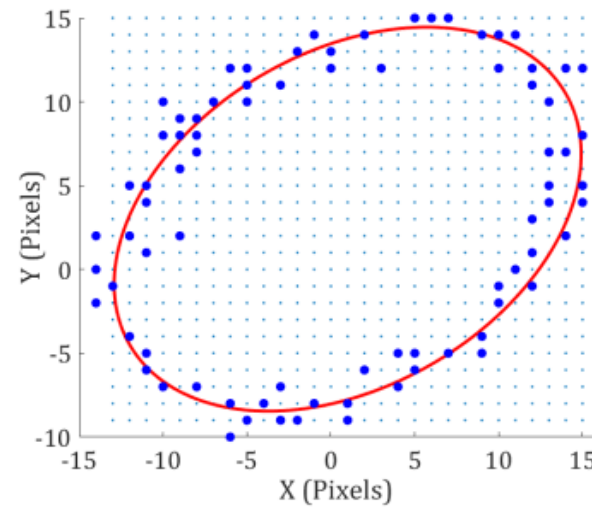


4.09 Ltr

upper band; lower band: unknown

How circumference come?

Least-Square Ellipse Fitting → 2nd Ramanujan's Formula for Ellipse Circumference



To create an ellipse that is closest to a point cloud

Approximation 3

Ramanujan also came up with this one. First we calculate "h":

$$h = \frac{(a-b)^2}{(a+b)^2}$$

Then use it here:

$$p \approx \pi(a+b) \left(1 + \frac{3h}{10 + \sqrt{4-3h}} \right)$$

It just an approximation tho

result

```
work/orange_shirt  4094 vs 4090  +0.1%
work/keng          2249 vs 2210  +1.8%
work/black_shirt   3648 vs 3510  +3.9%
work/sunshine      3093 vs 3130  -1.2%
work/champ         3354 vs 2000  +67.7% (truth disputed)
work/blue_shirt    5280 vs 3420  UNUSABLE
```

Keng:	upper	29.5	vs	28.61	-3.02%
	lower	19.5	vs	18.43	-5.49%
Sunshine:	upper	31	vs	33.2	+7.10%
	lower	22	vs	20.93	-4.89%
Champ:	upper	30	vs	31.13	+3.77%
	lower	20	vs	20.7	+3.60%

Ground truth on upper and lower band might diff of 5-10%
because we forget to measure at the same time of taking a photo

Project title

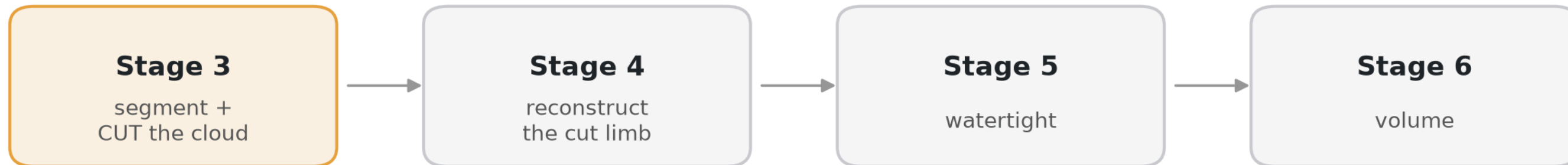
Champ: Few-shot 3D Reconstruction for leg measurements

Keng

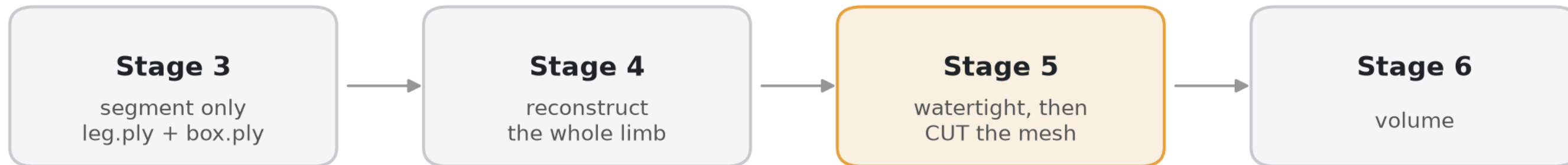
the cut moved to the mesh

Where the cut happens

BEFORE — Stage 3 cut



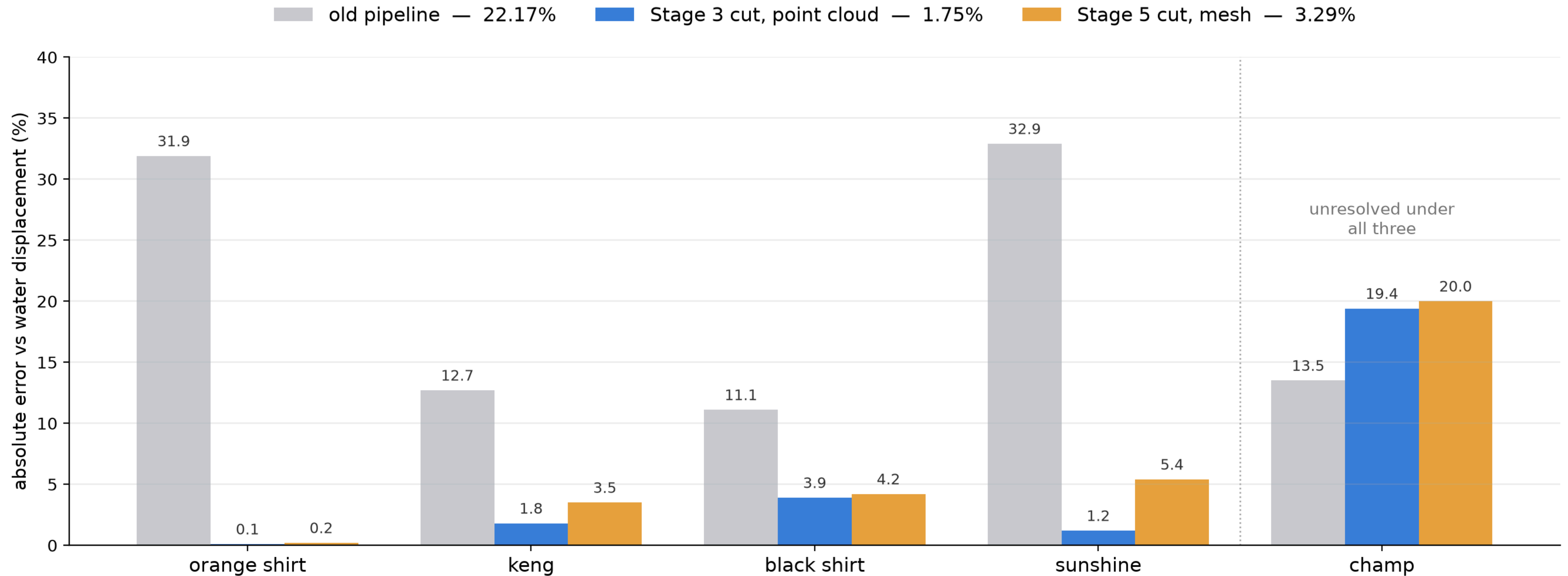
AFTER — Stage 5 cut



the cut now happens on a closed surface, so the person placing it sees a solid, not a scatter of points

- Why: a clinician places the cut. Placing it on a solid surface is far easier than on a scatter of points - that is the whole reason for the move.
- Re-cutting is now stages 5-6 only: a plane slice through geometry that already exists, about 7 s instead of re-running segmentation and Poisson.

Accuracy - three pipelines, same captures



old pipeline is the tree at bab2bbc: no Stage 0 framing, and 4 of 5 fell back to warp+floodfill, which leaks through an open surface. Its errors run both ways. Both current pipelines report watertight on every capture - so Stage 3 vs Stage 5 is the only clean comparison: only the cut moved.

What the mesh cut costs, and why

1.75% -> 3.29% mean absolute error

Every mesh-cut figure is HIGHER than its cloud-cut counterpart. None reads low.

- Stage 4 must now reconstruct the WHOLE limb, including the region above the cut: the far end of the capture, fewest views, most noise. The old order discarded it before Stage 4 ever saw it.
- sunshine carries most of the cost: Poisson returns $\chi = -2$ on its uncut limb, so Stage 4 falls back to an alpha shape that must open to 40x point spacing before it seals. A wrap that loose reads high. $-1.2\% \rightarrow +5.4\%$

Tried and rejected: Poisson depth 8/9/10, density trim 2/5/10%, largest-body selection.

The main body itself carries handles, so none of them produced a single closed solid.

Closing the limb's upper end before reconstruction recovered about a third: $+10.1\% \rightarrow +5.4\%$.

Place the cut on a photograph

The marker band is VISIBLE in the photo and only INFERRED in the mesh.

- The review can now show one of the frames VGGT actually consumed, with the limb drawn over it, and place the cut where you click the band.
- The camera comes from that frame's own extrinsic and intrinsic, with Stage 3's levelling rotation composed in. Checked against direct pixel projection: 0.15 px worst case, 0.04 px mean, over three frames.
- Field of view is taken PER FRAME - VGGT predicts a different focal for each (55.00, 54.40, 53.32 degrees on one capture), so it cannot be set once.

A click contributes the HEIGHT only; the limb's own axis gives the orientation.

Taking the tilt from the image too would be wrong: a line drawn on a photo back-projects to a plane through the camera centre, so the cut would tilt with wherever the photographer stood - and two people cutting the same band on different frames would get different volumes.

Also this session

Correctness

- Scene mesh was pairing the cube with the UNCUT limb while the number came from the cut one. Now re-merged from the cut limb.
- Colour was lost through the slice - trimesh drops it. Carried through by nearest vertex: saturation 0.000 -> 0.226, matching the uncut mesh's 0.223.
- Objects with no marker band had become unmeasurable. Fixed: nothing to cut is still an answer - the whole object is measured.

Housekeeping

- 1,213 lines of dead code deleted; everything that is a stage now lives under pipeline/.
- Every function documented: 105 missing docstrings -> 0, across Python and TypeScript.
- Samples regenerated so the demo is one pipeline throughout, not two.

Open

- sunshine's uncut reconstruction - fixing it would close most of the 1.75 -> 3.29 gap.
- champ still unresolved at ~20% under all three pipelines.